Functional Groups

Identifying, naming, properties and esters

Name _____ Class _____ Date _____

ENDINGS AND DATA

-ene (C=C) · -ol (–OH) · -al (–CHO) · -one (C=O in chain) · -oic acid (–COOH) · -oate (ester) · -amine (–NH_2) · -amide (–$CONH_2$) · chloro-, bromo- (halogens)

Molar masses (g/mol): C 12.01 · H 1.008 · O 16.00 · CH_3OH 32.04 · CH_3COOCH_3 74.08

Show units on every quantity in every step.

A. Understanding

1. (a) What is a functional group? (b) Why do ethanol and propan-1-ol have very similar chemical reactions?

B. Identifying and naming

2. Name the functional group in each compound: (a) CH_3CH_2OH (b) CH_3CHO (c) CH_3COCH_3 (d) CH_3COOH (e) CH_3COOCH_3 (f) $CH_3CH_2NH_2$ (g) CH_3CH_2Cl (h) CH_3CONH_2

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3. Name: (a) $CH_3CH_2CH_2OH$ (b) CH_3CH_2CHO (c) $CH_3COCH_2CH_3$ (d) $HCOOH$ (e) CH_3CH_2COOH (f) $CH_3CHClCH_3$

4. Write the condensed formula of: (a) butan-2-ol (b) pentanal (c) methyl propanoate (d) butanoic acid

C. Isomers and properties

5. (a) Draw condensed formulas for two compounds with the formula C_3H_6O that have different functional groups, and name them. (b) Do the same for C_2H_6O.

6. Boiling points: propane −42 °C, methoxymethane (CH_3OCH_3) −24 °C, ethanol 78 °C, ethanoic acid 118 °C. Explain this order in terms of the functional groups and the forces between molecules.

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D. Esters

7. (a) Name the ester formed from methanol and ethanoic acid, and write the equation. (b) Name the ester formed from ethanol and propanoic acid. (c) Name the acid and the alcohol that form propyl methanoate, $\mathrm{HCOOCH_2CH_2CH_3}$.

8. 4.00 g of methanol reacts with excess ethanoic acid: $\mathrm{CH_3COOH + CH_3OH \rightleftharpoons CH_3COOCH_3 + H_2O}$. (a) Calculate the theoretical yield of methyl ethanoate. (b) 6.50 g of ester is collected. Calculate the percent yield, and give one reason it is below 100 %.

E. Challenge

9. A sweet-smelling liquid contains 54.53 % C, 9.15 % H and 36.32 % O by mass, and its molar mass is 88.1 g/mol. (a) Find its empirical and molecular formulas. (b) It does not fizz with sodium carbonate solution, so it is not a carboxylic acid. Suggest which family it belongs to and give one possible name.

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